

Time Series Analysis – Final Project

Project Overview

The goal of this project is to perform a complete time series analysis on a real-world dataset. You are expected to explore the characteristics of the time series, identify its components and temporal dependence, apply smoothing and decomposition techniques, assess whether the remaining residuals behave like white noise, and, when necessary, build and evaluate an ARIMA or SARIMA model.

Your analysis should demonstrate not only that you can implement the required techniques, but also that you can interpret their results and justify your modeling decisions.

Important Principle: Justify Your Decisions

Do not treat this project as a sequence of code cells that must simply be executed. Throughout the project, explain what you observe and why you choose a particular method.

- Explain what you observe in the time series plot.
- Justify whether additive or multiplicative decomposition is more appropriate.
- Explain what the ACF and PACF suggest about temporal dependence.
- Explain whether the decomposition residuals can reasonably be considered white noise.
- Justify the ARIMA/SARIMA orders you select.
- Interpret model diagnostic results rather than reporting test statistics alone.
- Compare forecasting performance and explain which model you prefer.

Teams

This project is ideally intended to be done individually (I would recommend that). However, due to the tight schedule, you may choose to work in teams of 2. If you would like to register as a team, please fill out [this form](#). If you plan to work individually, you don't need to fill out the form. The form will be closed on Monday 17/8 at 11:59 PM.

Dataset

I will provide every team with the dataset to work with after the team registration closes.

1. Data Preparation

- Check the time index for duplicates, missing timestamps, and irregularities.
- Check for missing observations and explain how they are handled.
- Check for obvious data-quality issues and document any preprocessing.
- Ensure that the observations are correctly ordered chronologically.
- If a transformation is applied, explain why it is appropriate.

2. Exploratory Time Series Analysis

- Plot the original time series.
- Investigate the series for the components we discussed in the sessions.
- Use appropriate rolling statistics when useful.
- Use ACF and PACF plots to investigate temporal dependence.

3. Smoothing

- Apply at least two smoothing approaches covered in the course.
- Explain what the smoothing reveals about the underlying behavior of the series.
- Discuss any information that may be lost as a result of smoothing.

4. Time Series Decomposition

- Perform an appropriate decomposition of the series.
- Where appropriate, investigate both additive and multiplicative decomposition.
- Identify and interpret the trend, seasonal, and residual components.
- Explain which decomposition is more appropriate and justify your choice.

5. Decomposition Residual Analysis

- Analyze the residual component obtained from the decomposition.
- Determine whether the residuals can reasonably be considered white noise.
- Clearly explain the evidence supporting your conclusion.

6. ARIMA / SARIMA Modeling

- If the decomposition residuals are not adequately described as white noise, investigate an ARIMA or SARIMA model.
- Determine whether differencing is required and justify the choice.
- Determine appropriate non-seasonal orders p , d , and q .
- If seasonality is present, determine appropriate seasonal orders P , D , and Q and the seasonal period.
- Fit one or more candidate models.
- Use appropriate model-selection criteria.
- Explain why the final model was selected.

7. Model Diagnostics

- Analyze the residuals of the fitted ARIMA/SARIMA model.
- Determine whether the final residuals are reasonably consistent with white noise.
- If substantial structure remains in the residuals, reconsider the model rather than simply accepting the result.

8. Forecasting

- Split the dataset into training and test sets chronologically.
- Fit the selected model using the training data.

- Generate forecasts for the test period.
- Plot the actual observations, forecasts, and appropriate confidence intervals.
- Interpret the forecasting results.

9. Model Evaluation and Comparison

- Evaluate forecasting performance using appropriate metrics such as MAE and RMSE. MAPE may also be used when appropriate.
- If multiple ARIMA/SARIMA models are considered, compare their performance.
- Discuss whether the model with the best AIC/BIC is also the model with the best out-of-sample forecasting performance.
- Explain which model you would choose and why.

Expected Deliverable

A Jupyter Notebook containing the complete analysis. The notebook should be organized into clear sections and should contain both the code and written interpretation of the results.

- All code required to reproduce the analysis.
- All important plots and statistical test results.
- Written explanations and interpretations throughout the notebook.
- A final conclusion summarizing the analysis and selected forecasting model.

Note:

The quality of the project will be evaluated based on the quality of the reasoning as well as the correctness of the implementation. A technically correct model with little or no explanation will not receive full credit. Your analysis should tell a coherent story: what the data looks like, what temporal structure it contains, how that structure was modeled, whether the model is adequate, and how well it can forecast future observations.